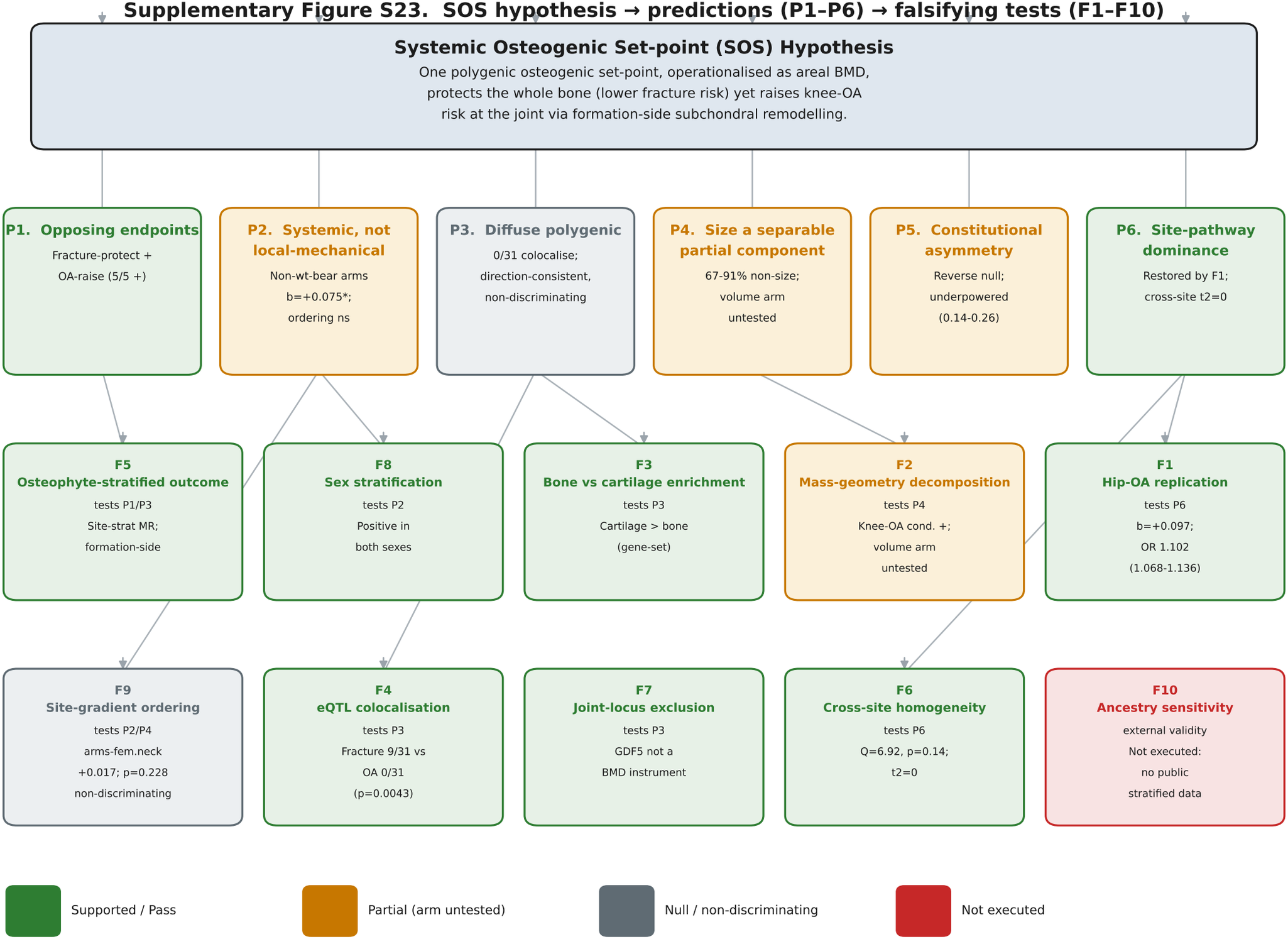
Supplementary Figure S23. SOS hypothesis → predictions (P1–P6) → falsifying tests (F1–F10)



Verdict: 9/10 executed tests do not refute SOS (F2 partial — volume arm untested; F9 non-discriminating; F10 unexecuted).
Kernel: one inherited variant set must fix opposite signs at whole-bone and joint endpoints; distinct variant sets or concordant signs refute it.
Priority closing tests: F2 volumetric arm (true vBMD instruments) and F10 cross-ancestry replication; P4 is discriminating and F2 decisive, with the volume arm unexecuted.
Supported, not demonstrated. * non-weight-bearing arms $\beta = +0.075$, $p = 1.7 \times 10^{-6}$ (systemic, not local-mechanical clause).